Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Engineering & Technology
B.Tech. | Sem-5 | Summer-2026

Course Code	:	2501CS501	**Date**	:	24-04-2026
Course Name	:	Building RESTful APIs with ASP.NET Core	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** What is an API? Write a short note on Key Characteristics of Web APIs. **4**

(B) Explain HTTP Request Components. **3**

OR

Explain HTTP Response Components.

(C) Create a Web API that inserts & deletes a record in given table. **7**
Student (ID, Name, Gender, Age, Department, City)

OR

Consume a Web API that gets all the record from API response & display it in web. Write an appropriate code for it.
Student (ID, Name, Gender, Age, Department, City)
End Point: localhost:8080/Student/GetAll

Q.2 **(A)** What is ORM? Explain Benefits of using an ORM. **4**

(B) Compare: Code First v/s Database First Approach. **3**

OR

Compare: DbContect v/s DbSet.

(C) What is LINQ Method Syntax? Explain LINQ Query & Method Syntax with an example. **7**

OR

Explain LINQ Projection Operators.

Q.3 **(A)** Define: API Versioning & Explain Major Update & New Features in API Versioning. **4**

(B) Write a note on following versioning types. **3**
1) URL Versioning
2) Header Versioning

OR

Write a note on Route Constraints.

	(C)	Explain Below Fluent Validation Methods. NotNull(), Equal(), Length(), InclusiveBetween(), LessThan(), Must() & Matches().	**7**
		OR	
		Explain Below terminologies in Fluent Validations. 1) Cascade Modes 2) Dependent Rules () 3) WithMessage () Method 4) WithName () Method	
Q.4	**(A)**	Compare: JavaScript v/s jQuery	**4**
	(B)	Explain Document Ready Event.	**3**
		OR	
		Explain the On () Method in jQuery.	
	(C)	Demonstrate jQuery Selectors with an example.	**7**
		OR	
		Demonstrate jQuery Form Events with an example.	
Q.5	**(A)**	Write a short note on Middleware and also explain why it is important.	**4**
	(B)	Compare: Authentication v/s Authorization.	**3**
		OR	
		Explain Common Built-in Middleware(s) available ASP.Net Core.	
	(C)	Demonstrate Role-Based Access Control in detail.	**7**
		OR	
		Demonstrate Action Filters in detail.	

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